COMP 511

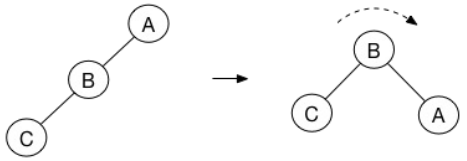
Computer Programming Foundations II

AVL tree (cont'd)

Recall Rebalance AVL tree

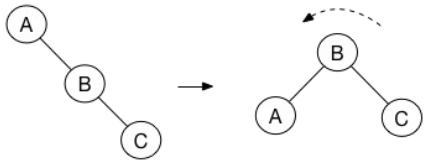
Rotations

- Single Rotation
  - Left Rotation (LL Rotation)



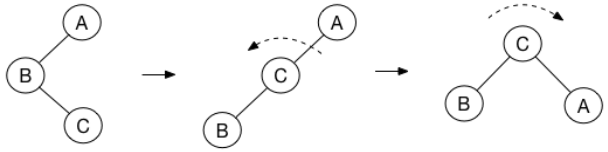
B becomes the new root.
A takes ownership of **B**'s right child as its left child (in this case, B's right child is null).
B takes ownership of **A** as its right child.

  - Right Rotation (RR Rotation)

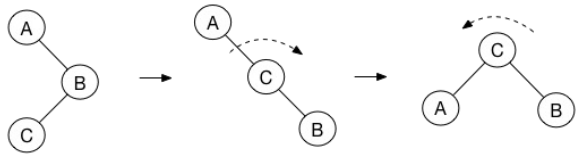


B becomes the new root.
A takes ownership of **B**'s left child, as its right child. (in this case, B's left child is null).
B takes ownership of **A**, as it's left child.

- Double Rotation
  - Left Right Rotation (LR Rotation)

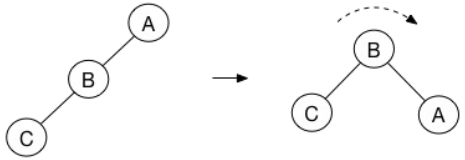


  - Right Left Rotation (RL Rotation)

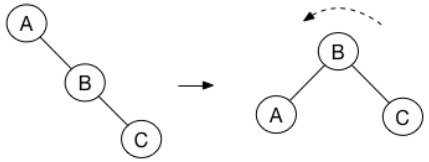


Rotations

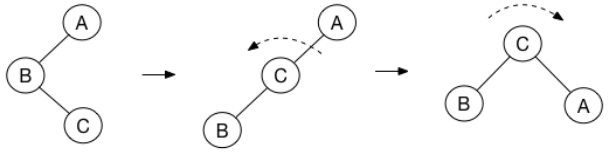
```
/**    (pseudocode)
 * Rotate binary tree node with left child.
 * For AVL trees, this is a single rotation for left/left. */
static BinaryNode rotateWithLeftChild( BinaryNode k2 )
{
    BinaryNode k1 = k2.left;
    k2.left = k1.right;
    k1.right = k2;
    return k1;
}
```



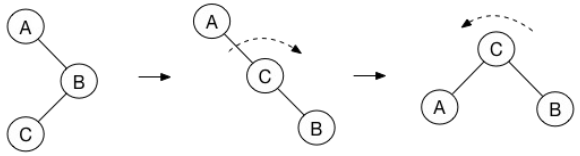

```
/** Rotate binary tree node with right child.
*  For AVL trees, this is a single rotation for right/right */
static BinaryNode rotateWithRightChild( BinaryNode N1 )
{
    BinaryNode N2 = N1.right;
    N1.right = N2.left;
    N2.left = N1;
    return N2;
}
```



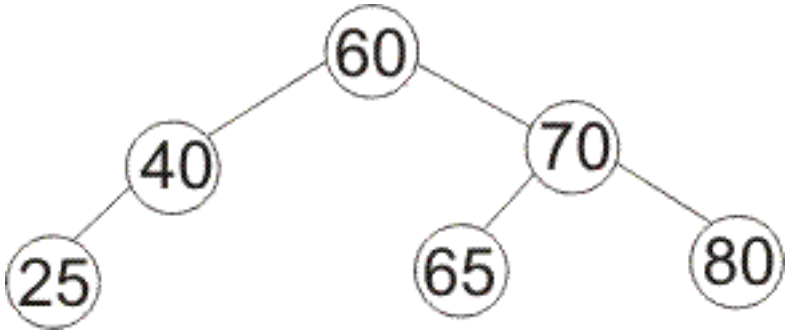

```
 /* Double rotations */
static BinaryNode doubleRotateWithLeftChild( BinaryNode k3 )
{
    k3.left = rotateWithRightChild( k3.left );
    return rotateWithLeftChild( k3 );
}
```




```
static BinaryNode doubleRotateWithRightChild( BinaryNode k3 )
{
    k3.right = rotateWithLeftChild( k3.right );
    return rotateWithRightChild( k3 );
}
```

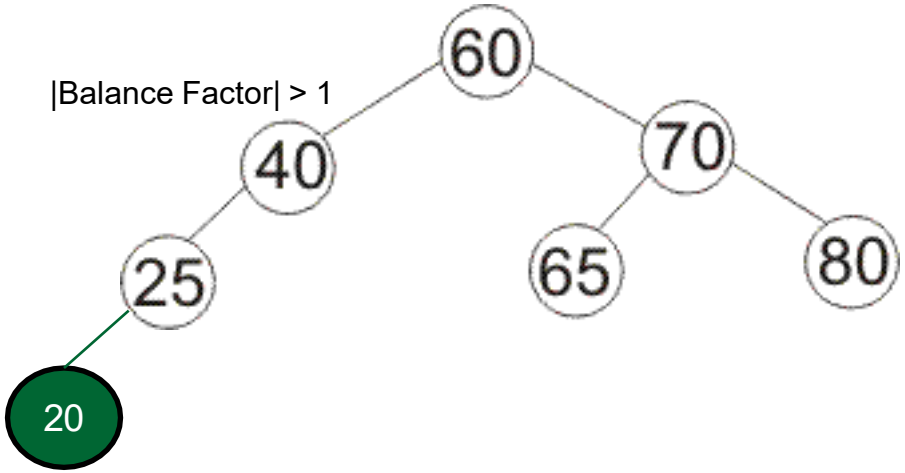


AVL Balance Example



Insert 20 into this BST and rebalance

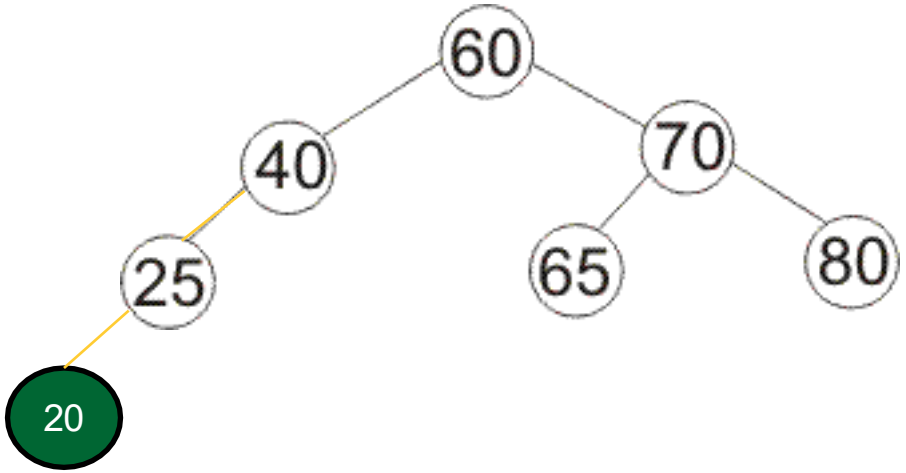
AVL Balance Example



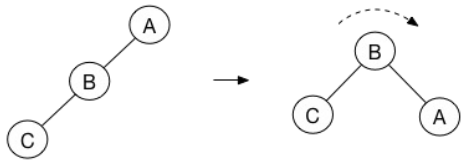
Insert 20 into this BST and rebalance

AVL Balance Example

```
static BinaryNode rotateWithLeftChild( BinaryNode k2 )
 {
    BinaryNode k1 = k2.left;
    k2.left = k1.right;
    k1.right = k2;
    return k1;
 }
```



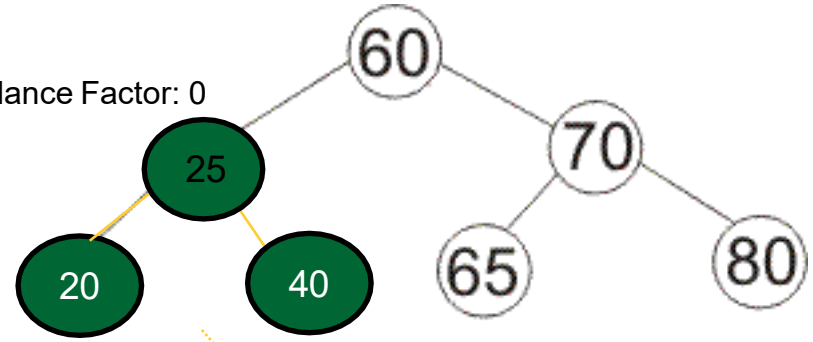
Recall from lecture:



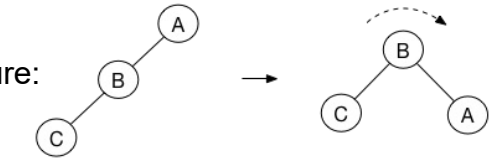
left child, left child rotation

AVL Balance Example

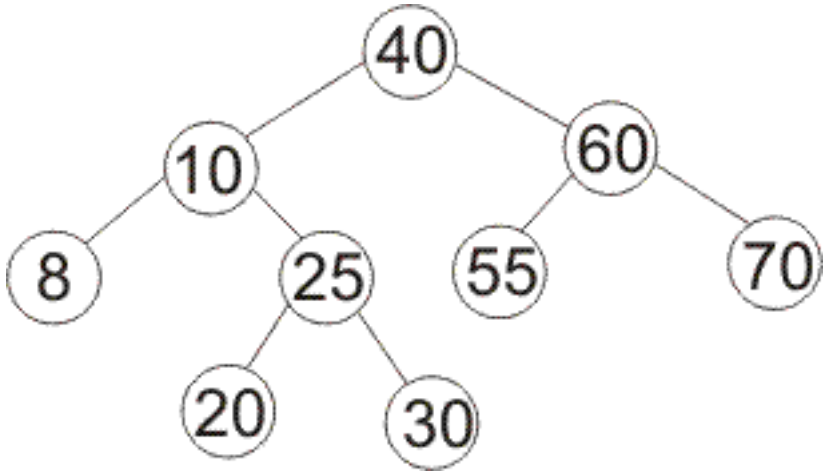
Balance Factor: 0



Recall from lecture:

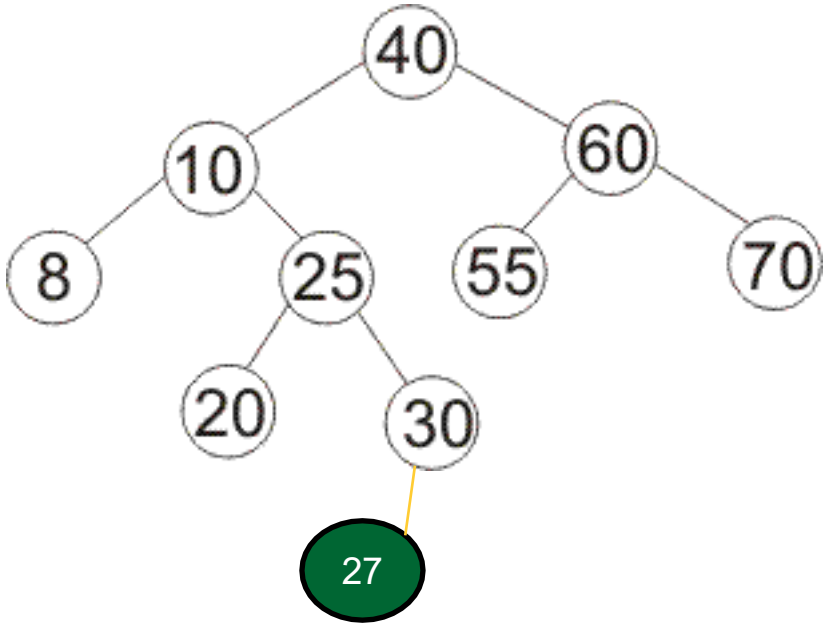


AVL Balance Example



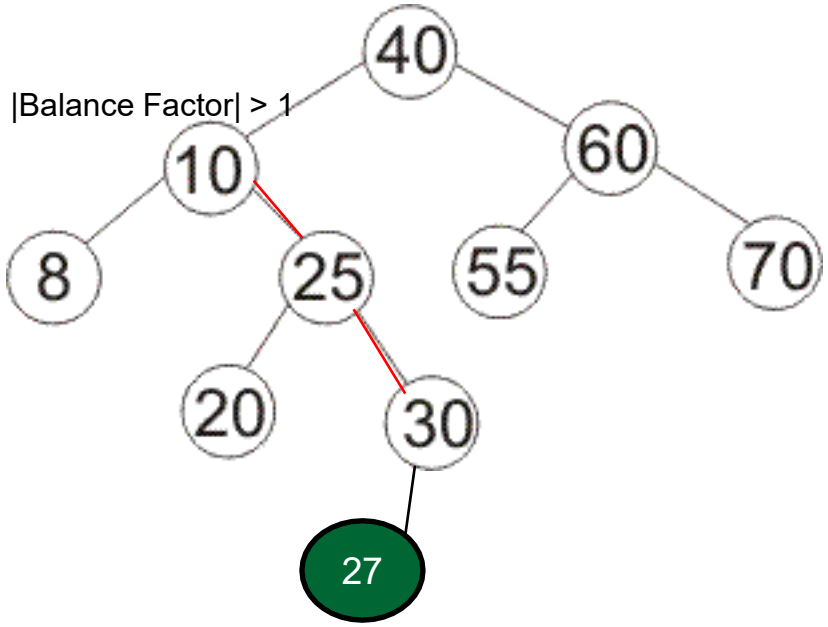
Insert 27 into this BST and rebalance

AVL Balance Example



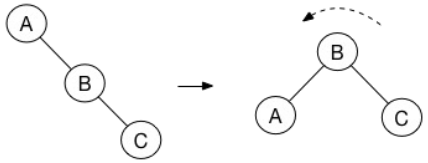
Insert 27 into this BST and rebalance

AVL Balance Example



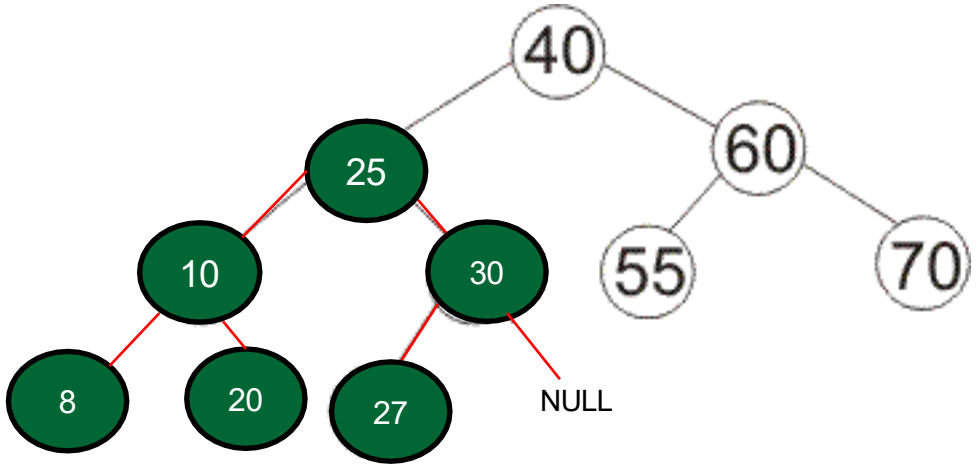

```
static BinaryNode rotateWithRightChild( BinaryNode N1 )
  {
     BinaryNode N2 = N1.right;
     N1.right = N2.left;
     N2.left = N1;
     return N2;
  }
```

Recall from lecture:

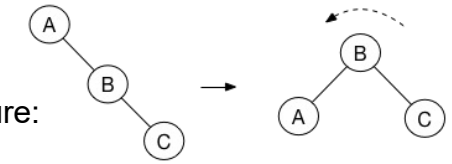


right child, right child rotation

AVL Balance Example

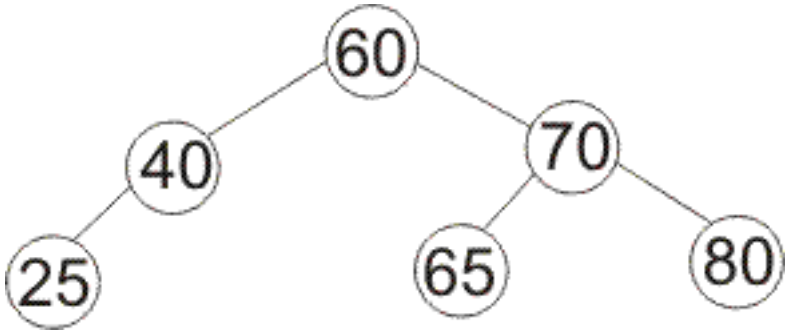


Recall from lecture:



right child, right child rotation

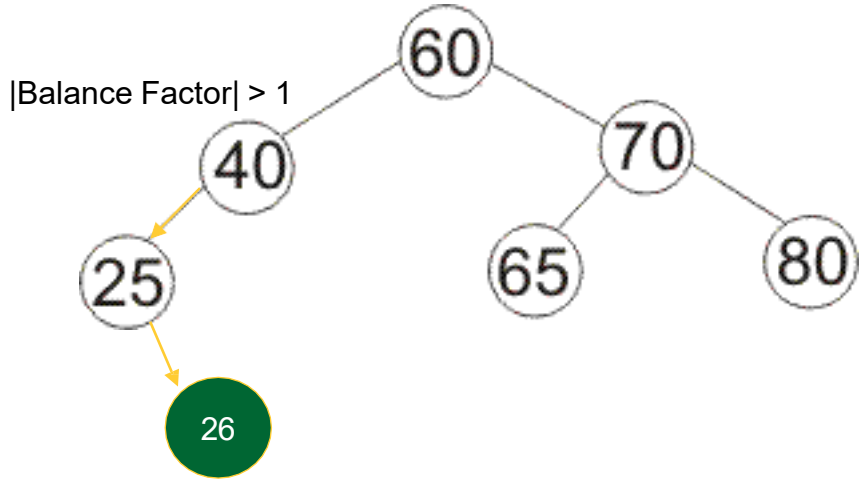
AVL Balance Example



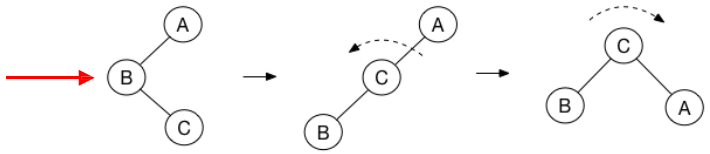
Insert 26 into this BST and rebalance

AVL Balance Example

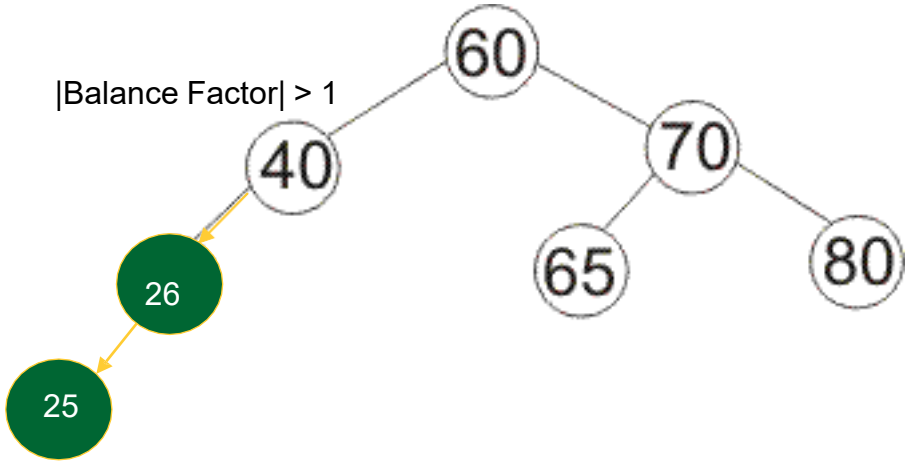
26 is inserted as a leaf



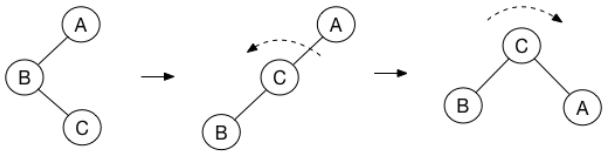
Recall from lecture notes:



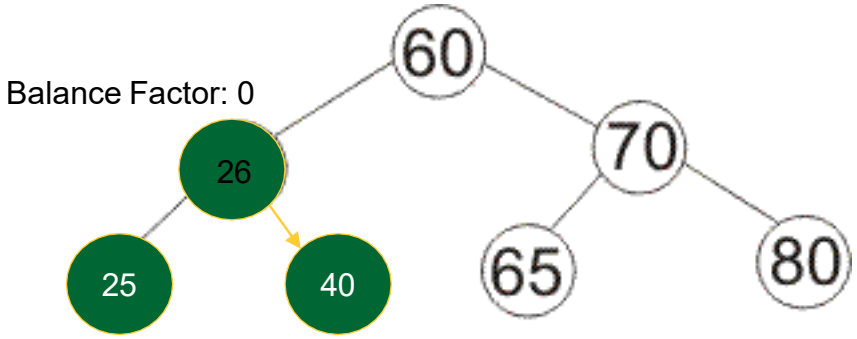
AVL Balance Example



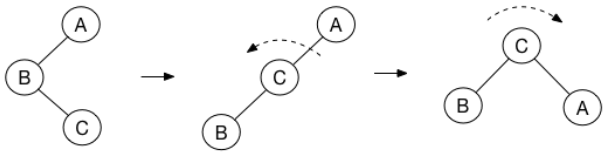
Recall from lecture notes:



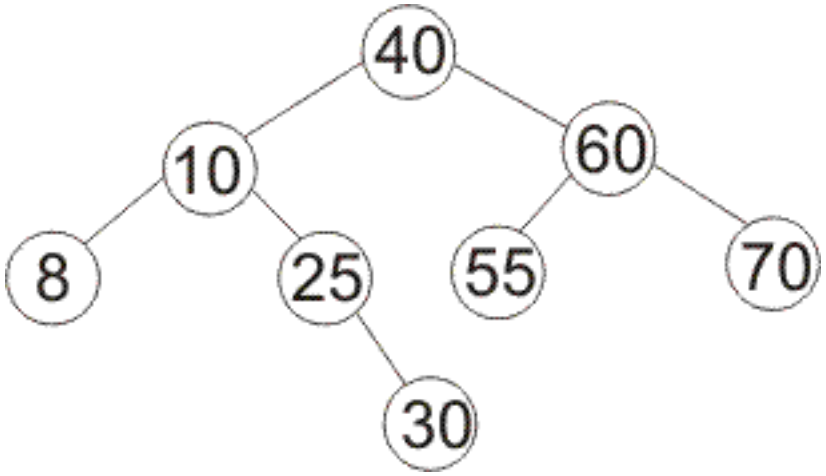
AVL Balance Example



Recall from lecture notes:

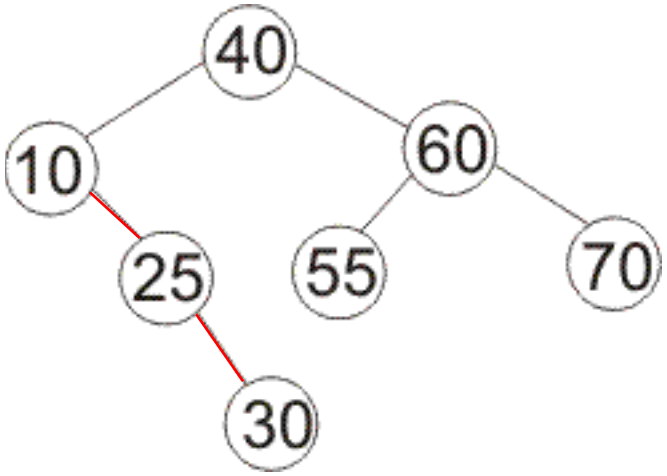


AVL Balance Example

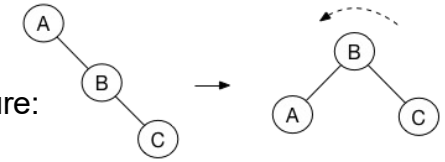


Remove 8 from this BST and rebalance

AVL Balance Example

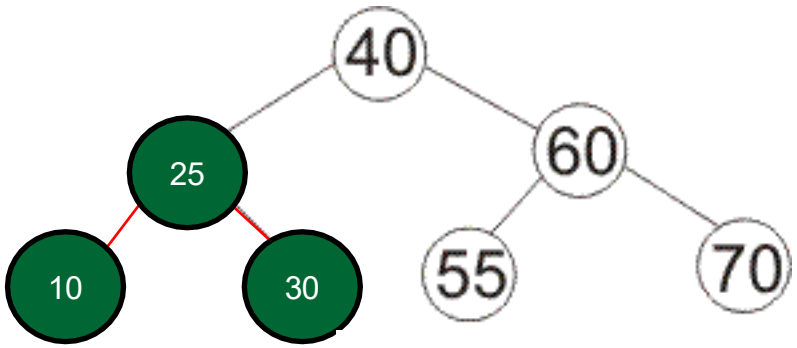


Recall from lecture:

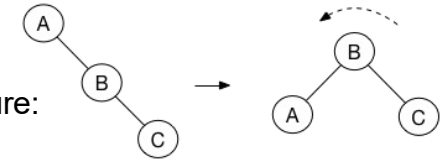


right child, right child rotation

AVL Balance Example



Recall from lecture:

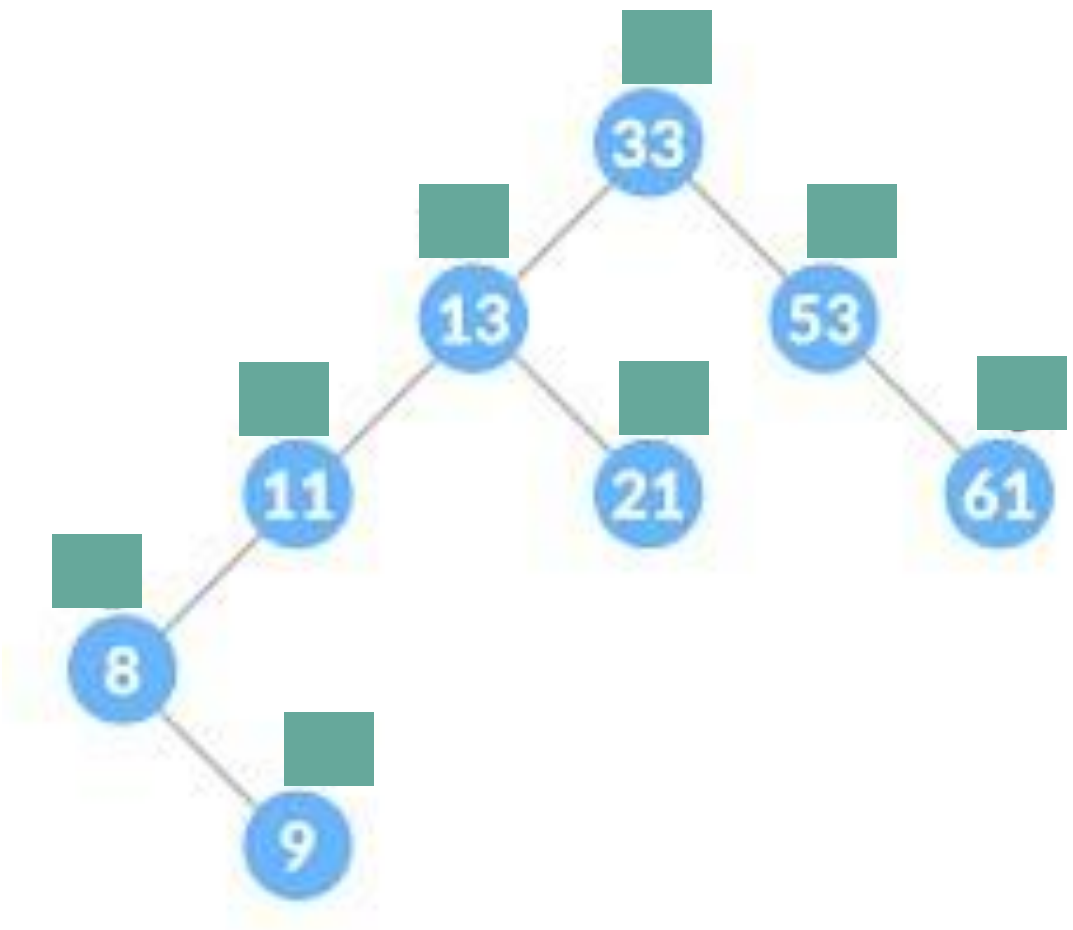


right child, right child rotation

Balance Factor

The difference between the AVL tree and the binary search tree is that the AVL balances itself after each operation that involves manipulating data.

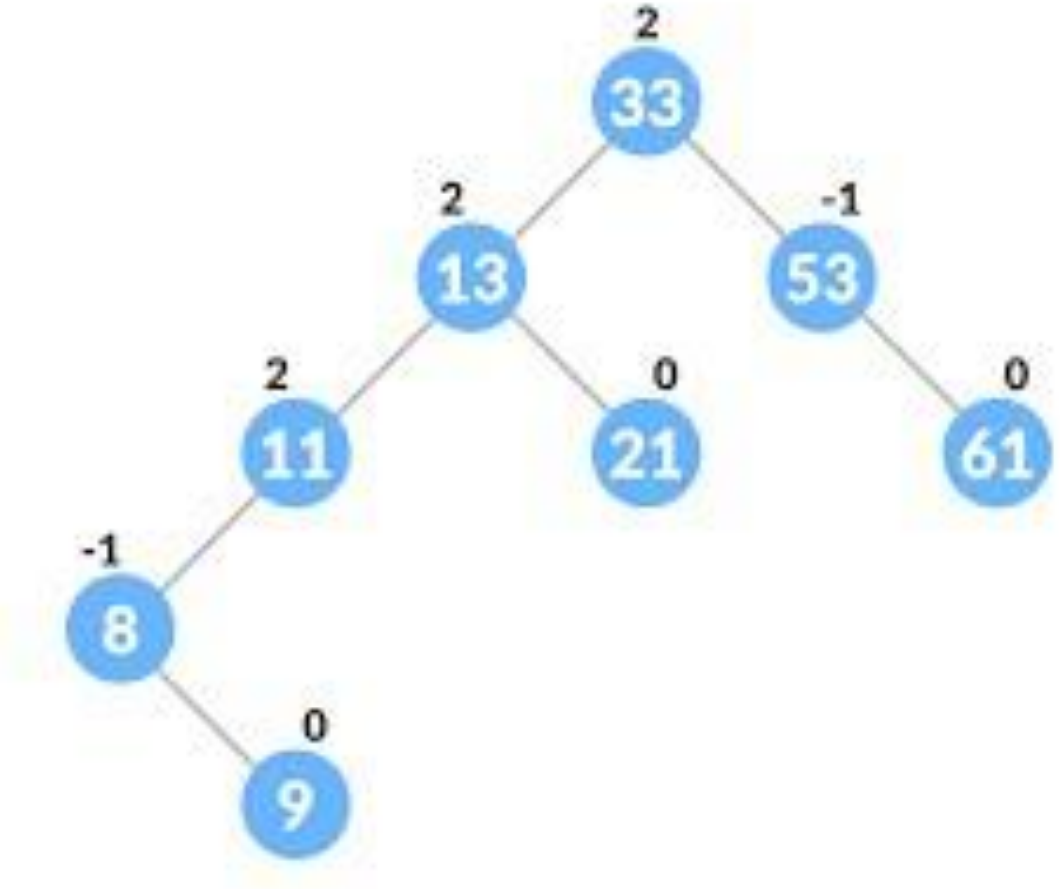
Using the formula: BalanceFactor(N) := Height(LeftSubtree(N)) – Height(RightSubtree(N))
Determine the balance factor of each node within the following BST.



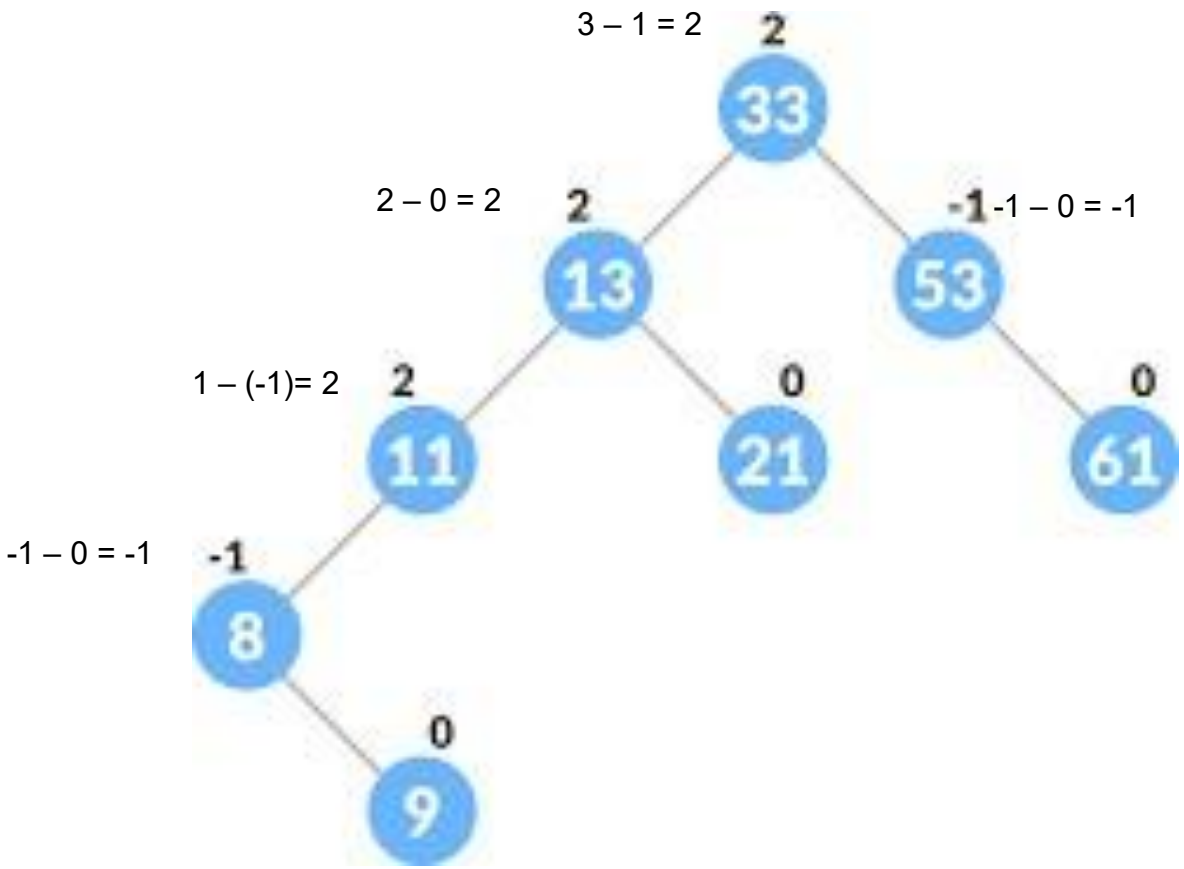
Balance Factor

The difference between the AVL tree and the binary search tree is that the AVL balances itself after each operation that involves manipulating data.

Using the formula: BalanceFactor(N) := Height(LeftSubtree(N)) – Height(RightSubtree(N))
Determine the balance factor of each node within the following BST.



AVL Balance Example



Create AVL tree

Create AVL tree

```
//pseudocode
public static void main(String[] args) {
        AVLtree tree = new AVLtree();

        for (int i = 1; i <= 10; i++)
            tree.insert(i); //includes rebalancing
}
```

Create AVL tree

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```

i: 1

root

```
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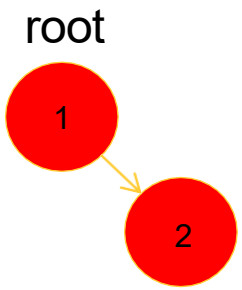
            tree.insert(i); //includes rebalancing
}
```

Create AVL tree

i: 2



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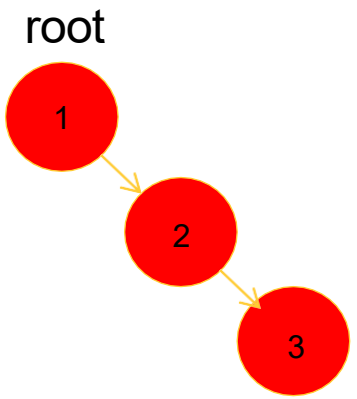
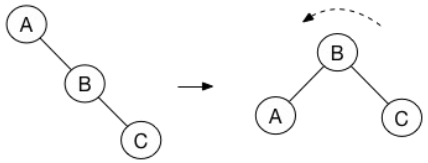
}
```

Create AVL tree

i: 3





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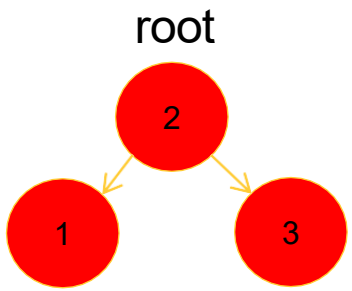
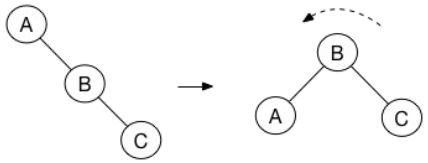
Create AVL tree

i: 3





```
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        AVLtree tree = new AVLtree();

        for (int i = 1; i <= 10; i++)
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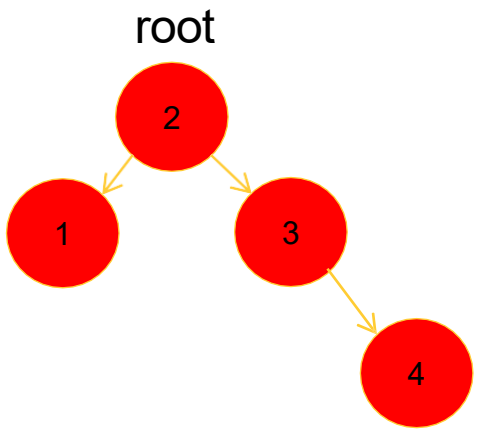
}
```

Create AVL tree

i: 4



```
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        AVLtree tree = new AVLtree();

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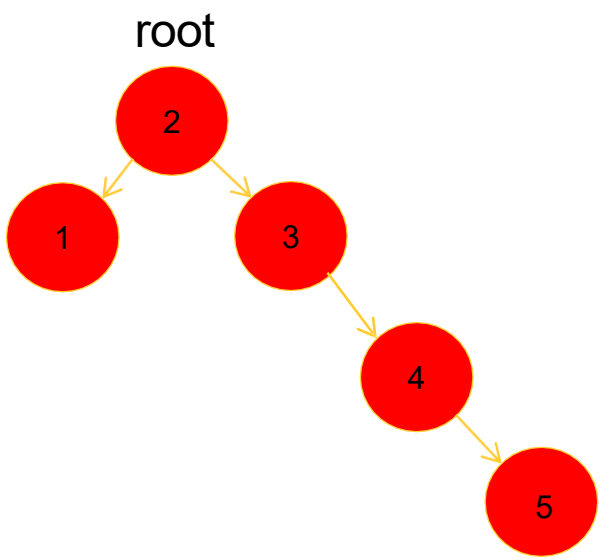
}
```

Create AVL tree

i: 5



In balance?

Create AVL tree

```
//pseudocode
public static void main(String[] args) {
        AVLtree tree = new AVLtree();

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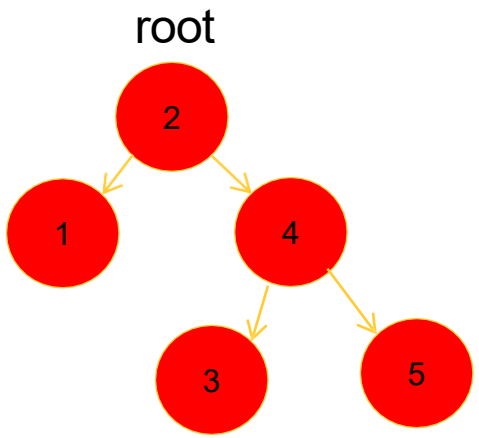
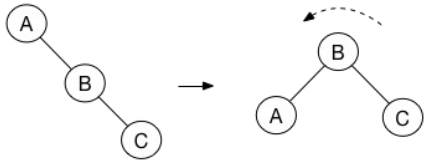
```

i: 5





Create AVL tree

```
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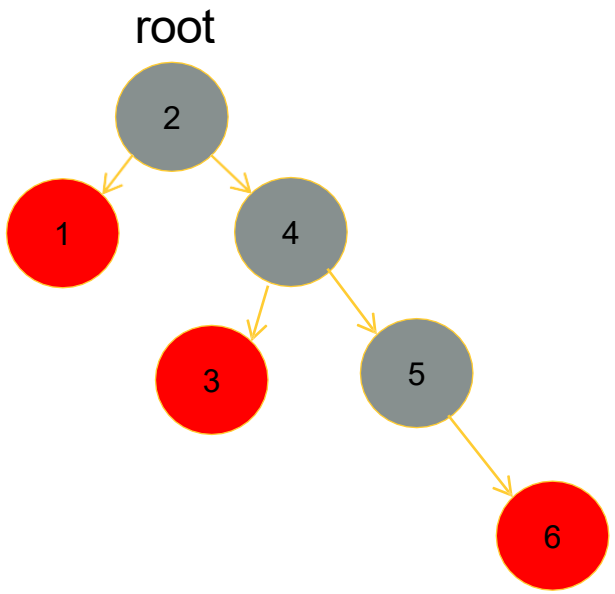
            tree.insert(i); //includes rebalancing
}
```

i: 6



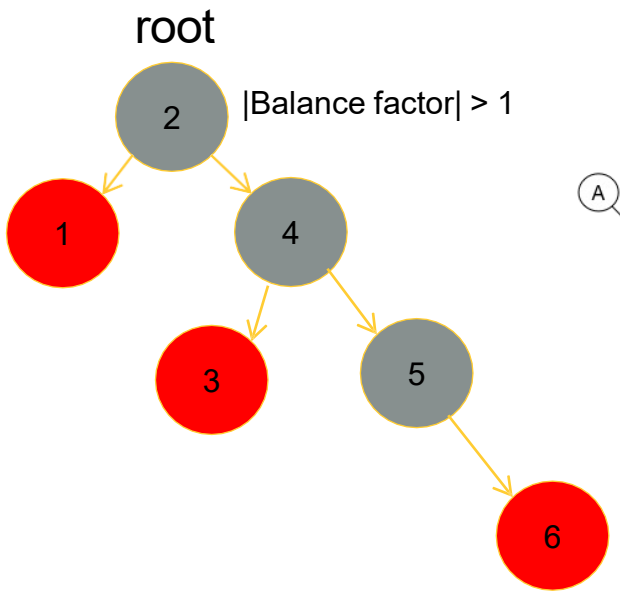
In balance?

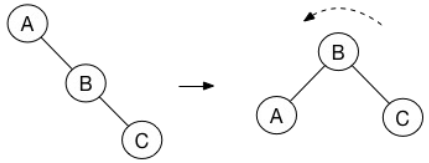
```
//pseudocode
public static void main(String[] args) {
        AVLtree tree = new AVLtree();

        for (int i = 1; i <= 10; i++)
            tree.insert(i); //includes rebalancing
}
```

Create AVL tree

i: 6



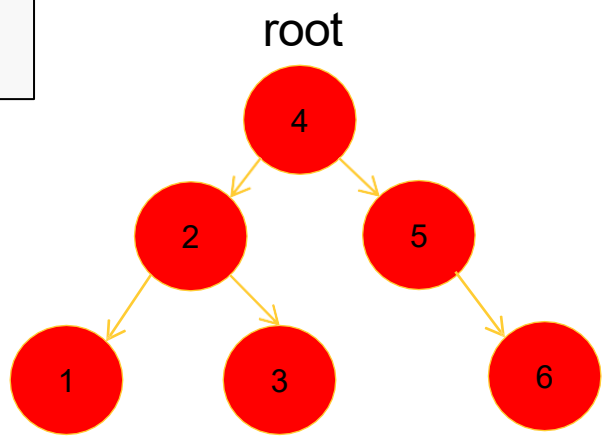


Create AVL tree

```
//pseudocode
public static void main(String[] args) {
        AVLtree tree = new AVLtree();

        for (int i = 1; i <= 10; i++)
            tree.insert(i); //includes rebalancing
}
```

i: 6

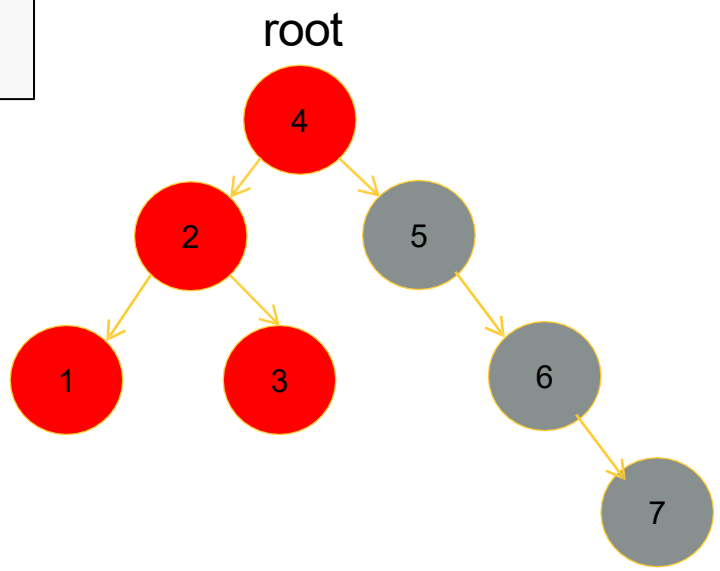


Create AVL tree

```
//pseudocode
public static void main(String[] args) {
        AVLtree tree = new AVLtree();

        for (int i = 1; i <= 10; i++)
            tree.insert(i); //includes rebalancing
}
```

i: 7

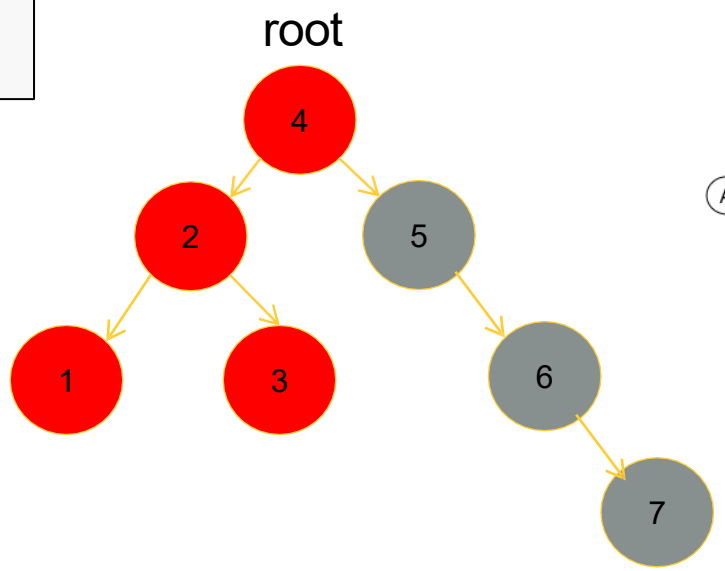


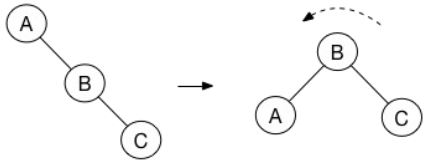
Create AVL tree

```
//pseudocode
public static void main(String[] args) {
        AVLtree tree = new AVLtree();

        for (int i = 1; i <= 10; i++)
            tree.insert(i); //includes rebalancing
}
```

i: 7





Create AVL tree

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//pseudocode
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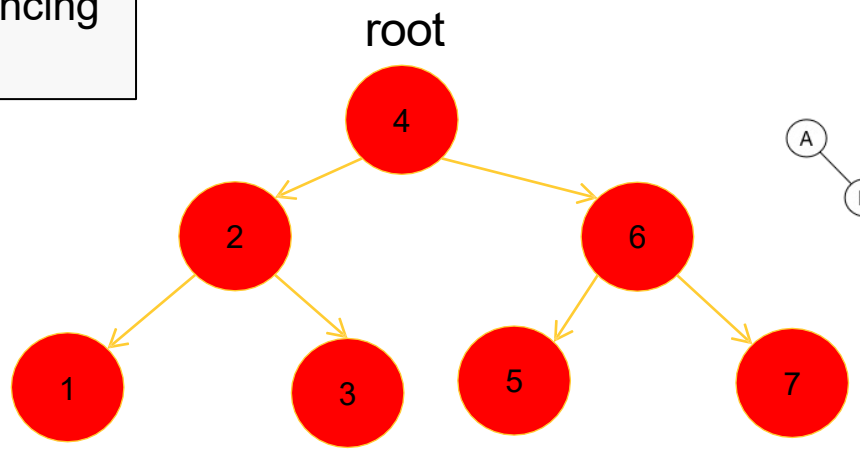
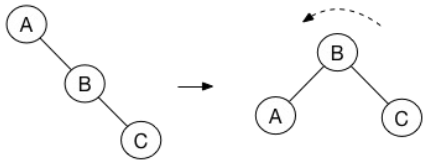
        AVLtree tree = new AVLtree();

        for (int i = 1; i <= 10; i++)
            tree.insert(i); //includes rebalancing
}
```

i: 7



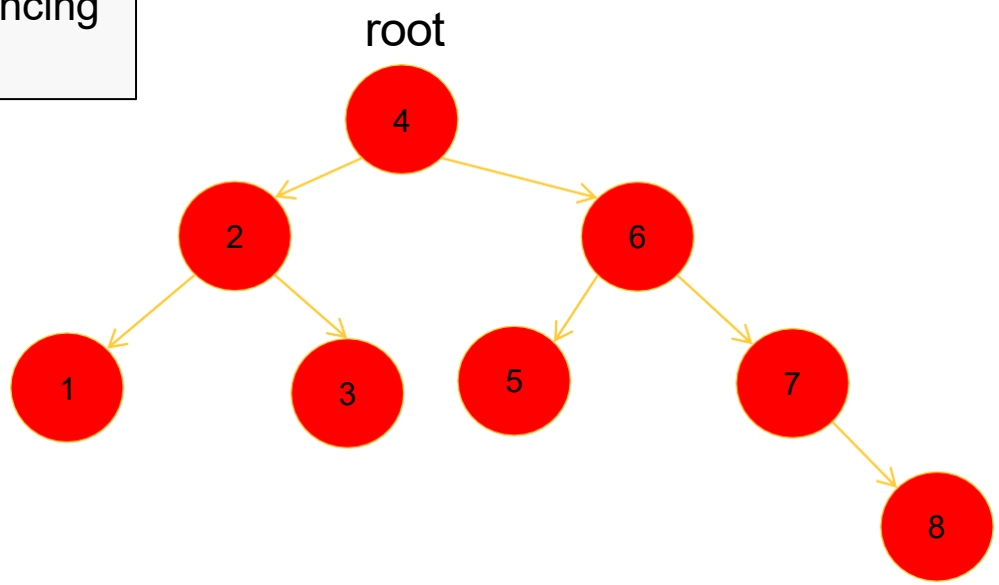


Create AVL tree

```
//pseudocode
public static void main(String[] args) {
        AVLtree tree = new AVLtree();

        for (int i = 1; i <= 10; i++)
            tree.insert(i); //includes rebalancing
}
```

i: 8

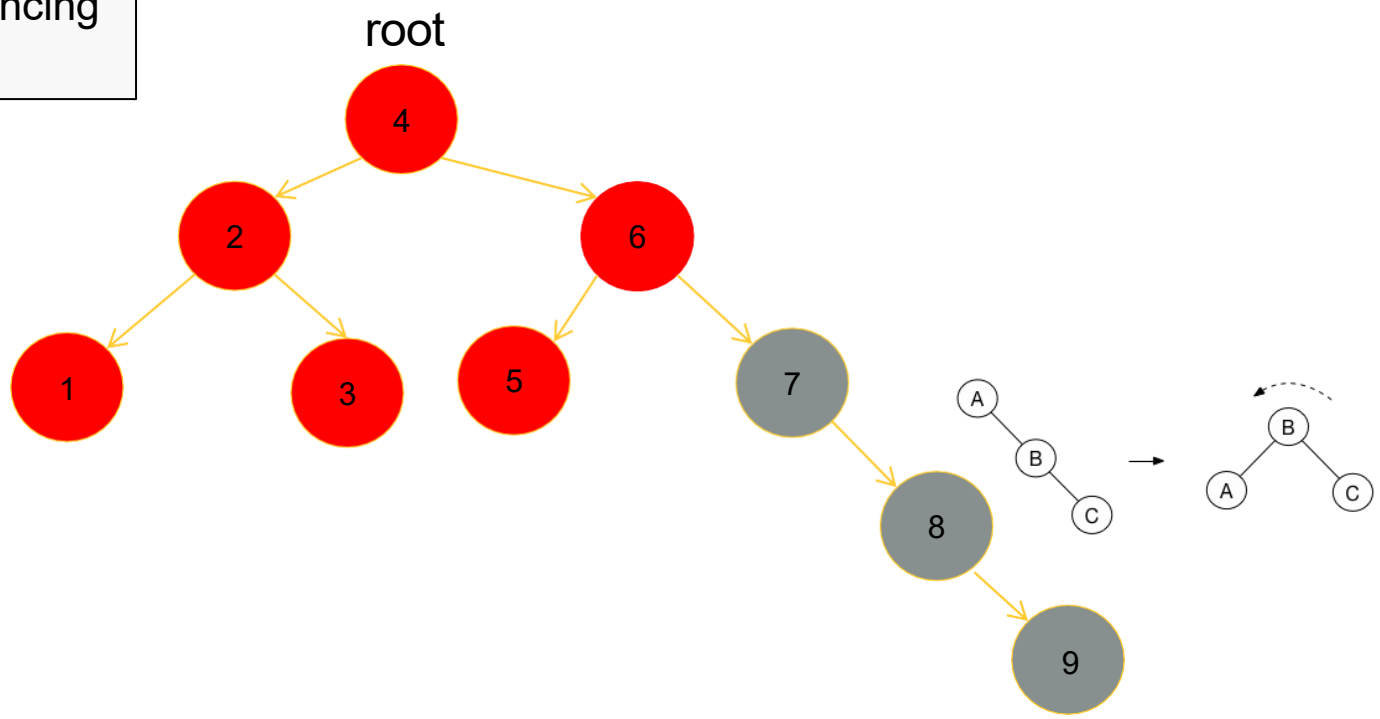


Create AVL tree

```
//pseudocode
public static void main(String[] args) {
        AVLtree tree = new AVLtree();

        for (int i = 1; i <= 10; i++)
            tree.insert(i); //includes rebalancing
}
```

i: 9

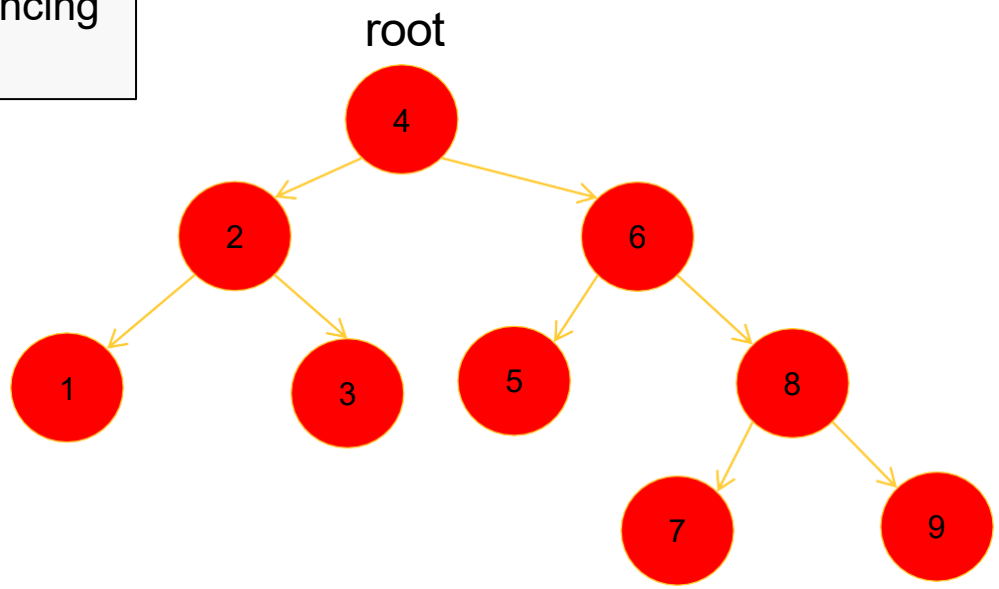


Create AVL tree

```
//pseudocode
public static void main(String[] args) {
        AVLtree tree = new AVLtree();

        for (int i = 1; i <= 10; i++)
            tree.insert(i); //includes rebalancing
}
```

i: 9

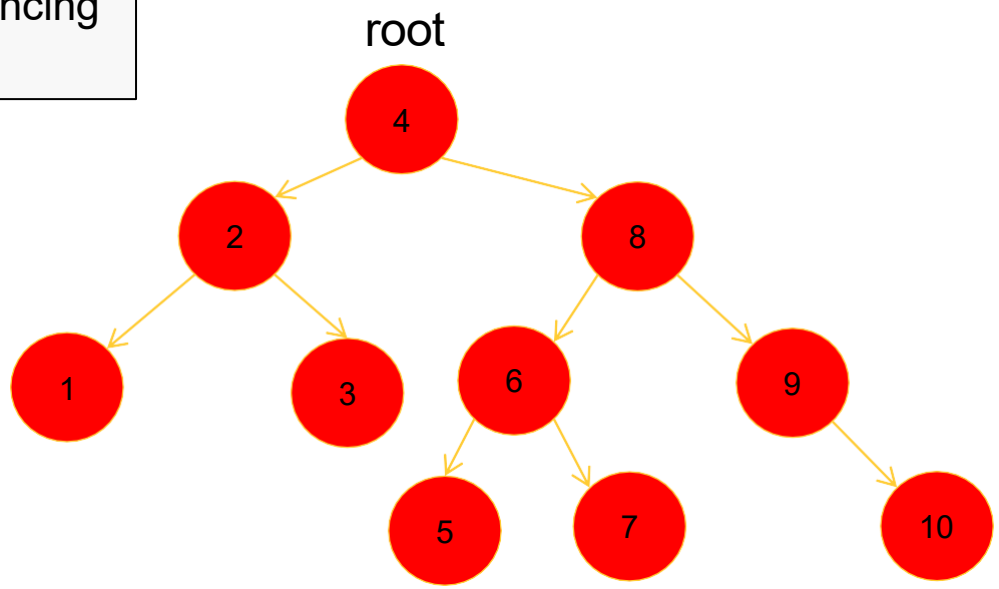


Create AVL tree

```
//pseudocode
public static void main(String[] args) {
        AVLtree tree = new AVLtree();

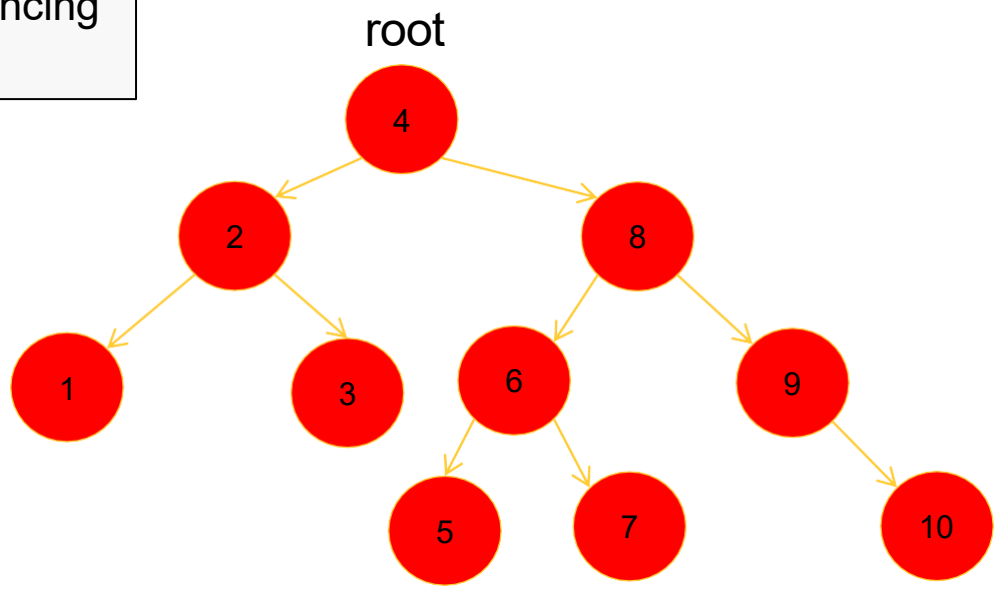
        for (int i = 1; i <= 10; i++)
            tree.insert(i); //includes rebalancing
}
```

i: 10



Create AVL tree

```
//pseudocode
public static void main(String[] args) {
        AVLtree tree = new AVLtree();

        for (int i = 1; i <= 10; i++)
            tree.insert(i); //includes rebalancing
}
```

i: 10